

Ekaansh Patel

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EDUCATION

University of Michigan, College of Literature, Science, and The Arts

Expected May 2028

B.S. in Computer Science, GPA: 3.6

Ann Arbor, MI

Awards: Fall 2024 and Fall 2025 University Honors List

Coursework: Data Structures and Algorithms, Computer Science Pragmatics (Developer Tools and Ecosystems), Introduction to Computer Architecture, Foundations of Computer Science (Computational Complexity and Computability), Introduction to Machine Learning (Fall 2026), Web Systems (Fall 2026)

TECHNICAL SKILLS

Languages: C++, Python, HTML/CSS, TypeScript, JavaScript, Java

Libraries/Frameworks: Next.js, React, Tailwind CSS, Playwright, Pandas, NumPy

Developer Tools: VS Code, Git, Linux/Unix, npm, Vite

PROJECTS

Formula 1 Race Analytics Platform

- Engineered a predictive Formula 1 dashboard using React, JavaScript, Vite, and Tailwind CSS, to simulate and visualize race progressions based on predictive modeling.
- Developed a robust data processing pipeline in Python, utilizing Pandas and NumPy to parse and aggregate massive race datasets via the FastF1 and OpenF1 APIs.
- Designed a structured data integration layer to feed processed Python model outputs directly into the React frontend, optimizing state management for real-time interval metrics.

Drone Delivery

- Engineered a Minimum Spanning Tree utilizing Prim's algorithm to optimize network connectivity across all drone delivery locations.
- Developed a fast heuristic Traveling Salesperson Problem (TSP) solver via the Farthest Insertion algorithm to rapidly approximate efficient flight paths.
- Implemented an optimal TSP solver using a Branch and Bound algorithm, guaranteeing the absolute shortest delivery route by aggressively pruning suboptimal paths.
- Built an interactive web dashboard using HTML, JavaScript, and CSS to visualize complex map data, specific delivery locations, and exact drone flight paths.

Machine Learning - Classifier

- Built a text classification system to automatically identify the subject of course discussion board posts based on post content.
- Applied NLP and machine learning techniques using efficient data structures (sets and maps) to extract and manage word and label statistics.
- Trained the model on historical Piazza posts and implemented a log-likelihood-based classifier to predict the most probable label for each post.

Text Editor

- Created a basic text editor where you are able to type, delete, and move around between text and different lines of text
- Uses abstract data types, dynamic memory, iterators, and linked lists for efficiency in order to move around the different text data

PROJECT TEAMS & ACTIVITIES

Michigan Mars Rover Team (Mrover) | Teleoperations Software Subteam

September 2025 – Present

- Engineered responsive UI features for the online base station, implementing Playwright end-to-end test suites to validate functionality and prevent regressions.
- Enhanced the operator experience by redesigning the data interface, transitioning from legacy data tables to intuitive, clickable components mapped directly onto a visual model of the rover.

Wolverine Sports Analytics

September 2025 – Present

- Collected and scraped player statistics from multiple sports websites, normalized the data into CSV datasets, and implemented a fantasy football projection engine that estimates waiver-wire player point totals using a mathematical model incorporating player usage and historical performance.